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Thermoelectric integration engineer

Thermoelectric cooling integration across CPU thermal management and battery thermal management systems. Six-generation iterative development documented publicly under handles K0paz / Nopaz / KopaZ. Results documented across two operating regimes: daily-use (passive radiator hot-side rejection, DI water) and benchmark-regime (1 HP aquarium / bathtub chiller on hot side). Cold path is thermoelectric in both regimes. Lowest recorded coolant reservoir temperature (post-CPU): −19.125 °C.

Projects

Thermoelectric CPU cooling — 12× Laird UTX6 stack (2023 — present)

TEC integration for CPU thermal management at sub-ambient and sub-zero coolant temperatures. 12× Laird UTX6-24-F1-5555 modules clamped to three custom aluminum waterblocks (Fusion 360), run derated at 39–46% of Imax so Peltier cooling stays ahead of Joule heating, cooling a direct-die consumer CPU through liquid metal.

Two independent hydronic loops: DI-water hot side rejecting through dual 360 mm push-pull radiators (chiller-assisted for benchmark runs), and a sub-zero cold side on a positive-displacement pump. Coolant chemistry worked out across ethylene glycol, propylene glycol and propanediol-1,3 blends to hold viscosity down at the freeze point the loop actually reaches; pump chosen for displacement against that viscosity. TIMs selected per interface: graphite sheet and silicone paste at the TEC/block joint, liquid metal at the die. Condensation handled by insulating lines, reservoir and interfaces and conformal-coating the PCB.

Instrumented with a DS18B20 probe in the cold-side reservoir (post-CPU), thermal imaging on the radiator hot side, and V/I readings from the buck-boost converters feeding each 6-module bank to set TEC input power. Coolant temperature is tied to CPU telemetry (clock, package power, voltage, cycle counts) through a custom PuTTY capture script alongside HWMonitor/OCCT, with Excel and Python for analysis; logs are public. From that data: COP 0.43 at ΔT 26.8 K, stack thermal resistance ~0.18 °C/W, −19.125 °C coolant post-CPU with chiller-assisted hot side, 3.56 °C at 285 W input in daily use.

Six hardware iterations: V1 8× TEC1-12706 (~100 W) → V4 12× TEC2-25408 two-stage (~400 W), where Qc collapsed under load and the COP–ΔT analysis pushed the design toward fewer, derated industrial modules → V5 4× Laird ETX25 (~300 W, validated on a 70.6 mm² die) → V6 the current 12× UTX6 stack. Liquid-metal failure modes below 0 °C documented along the way.

Routine upkeep: re-applying TIMs after thermal cycling, cross-checking instruments, coolant top-off and periodic changes, and loop modifications (module count/type, waterblock service). Results published as IEEE-style writeups and forum reports; benchmark entries carry full sensor telemetry.

Measurement: AI (Claude) Coded sensorpoint.bin extraction program

Gated process that boots at PC startup, with a hardware-selection category (RAM, CPU, GPU, storage, motherboard) choosing which sensorpoints to extract. Rows are fused with coolant reservoir tank temperature data for CPU VFT curve / stability evaluation. Sensor data is saved to local storage as .csv and pushed to GitHub by the user.
dongk99/CPU_DATA: AMD R7 9850X3D CPU Data, Subzero cooled with TECs. See TECloopLogs for methodologies.
https://github.com/dongk99/CPU_DATA

Benchmark results

HWBot Cinebench R23 Single Core (BenchMate), Ryzen 7 9800X3D: #5 of 64 users; the four above use LN2 or phase change. OCBase Single Thread SSE leaderboard, 9800X3D: ranks #1–64 under KopaZ as one systematic V-F-thermal sweep with full HWINFO telemetry per run (SSE chosen to keep RAM/FCLK/IMC out of the measurement); 9850X3D board also held.
https://hwbot.org/benchmarks/cinebench_-_r23_single_core_with_benchmate/rankings?processorId=7463&hardwareType=cpu
https://www.ocbase.com/benchmark

Documentation

Technical writeup (V5, 4× ETX25, direct-die):
https://www.reddit.com/r/overclocking/comments/1kk901d/9800x3d_cinebench_results_with_new_tec_setup/
Liquid-metal failure-mode report, sub-zero direct-die:
https://reddit.com/r/overclocking/comments/1ocvi85/
HWBot profile:
https://hwbot.org/users/nopaz/
Live rig footage (41 unedited livestreams under load):
https://www.youtube.com/@kopazwashere
Raw coolant temperature logs and capture scripts:
https://github.com/dongk99/TECloopLogs

LLM agent harness: RimWorld “VerbalCommands” (2026 – present) · github.com/dongk99/RimworldVerbalCommands

- C# integration layer (Harmony mod, 73 source files) that lets LLM agents control a live game through plain-English orders, with no screen capture or MCP.
- One backend runs the Anthropic Messages API, any OpenAI-compatible server (OpenAI, Ollama, LM Studio, OpenRouter, Gemini, vLLM), or Claude Code headless. Request and tool-call formats are translated between providers.
- *Claude code harness (current testing harness) for Agent-play feature.
- Reliability:
  - Rate-limit and overload errors are mapped to plain messages, with a single fallback retry.
  - Inbox and outbox files are written atomically (temp file, then rename).
  - Event-stream state updates, stall detection, and logging of every order.
- Wrote my own sentence router (lexer, parser, compiler, runner; 2,274 lines of rules). Most orders go through a small model (Haiku), and unclear ones come back as a question instead of a guess. Agent-play models are tested with Sonnet 5.5, Opus 5.5, Fable 5.1 and Haiku 5.5.
- Multi-agent orchestration: specialist agents act on their own for reversible actions. Actions that use up resources go to the orchestrator as written proposals, which it approves, denies or edits.
- Evaluation against a human reference on the same save: the unsupervised agent colony reached 15,159 wealth at day 10.5, against 24,129 for the human play at day 10. Tokens are metered per order. Qualitative-Quantitative eval based on poor judgement (e.g. assigning a bad colonist to do a job that they aren't fit for), which is intended as an inference test as a real-life HR/work assignment task.
- API keys never appear in the UI and are stored outside the mod folder, and each provider’s key is sent only to that provider.

TEC-controlled BTMS — Electric motorcycle battery (2019 — present)

Three battery iterations

72 V 150 Ah (abandoned — impractical dimension / weight for frame mounting).
72 V 45 Ah (1s9p × 20 series, in service).
72 V 60 Ah (1s12p × 20 series, current BTMS testbed; manufacturing-aware single-axis CNC spot-welding design).

BTMS

4×2 liquid-cooled TEC1-12706, current-limited to ~20 % Imax (9.6 A).
Calculated COP 2.5–2.7 at ΔT 10 K; ~2.0 with circulation losses.
Thermistor feedback via microcontroller with PID control on highest-recorded cell temperature (FPGA replacement planned).
TECs selected over vapor-compression for dimension, absence of moving parts, bidirectional capability.

Fusion 360 assemblies

V4 bracket, 4× ETX25 TEC:
https://a360.co/4eCBjGM
V4 bracket, improved:
https://a360.co/4vIZSIa
Experimental direct-coolant design (fin clamped to TEC face, touching coolant):
https://a360.co/4t9A5Xr

120×160 waterblock clamp (part level):
https://a360.co/4eCBou4
Custom battery bracket, flush cathode / anode thermal interfacing:
https://a360.co/4sU8l8S

Personal eVTOL design (in progress — structural battery integration)

Integration target: structural battery packs as load-bearing airframe elements, eliminating separate battery enclosures.

Fusion 360

Full assembly (in progress):
https://a360.co/4cOMFoS
Battery subassembly:
https://a360.co/4u1qvX4

3D Interactive Periodic Table Viewer (2024 — present)

Repository:
https://github.com/dongk99/3D_IPV
118 elements, 32 properties, 15 plottable axes, 29 toggleable tooltip filters.
Adaptive log / linear normalization (threshold at 2.2 orders of magnitude dynamic range).
Handles 7.8 orders of magnitude spread across properties without visual collapse.

Primary references for dataset

IUPAC 2021–2024, NIST ASD, CRC Handbook, Pyykkö & Atsumi covalent radii, Schwerdtfeger & Nagle 2025 polarizabilities, Clementi-Raimondi effective nuclear charges, Desclaux / Tatewaki relativistic contraction ratios.
20 Madelung-exception ground-state configurations handled experimentally, not predicted.
CC BY 4.0 dataset, MIT code.

Custom 3D Modeled/printed drones

Relevant work experience: Senior drone repair tech, 2021~2022, DDS (DJI Drone Service), 2019~2020
Designed FFF Printable drone frames using a 300x300 3D printer, Rapid prototyping in ~3 month to finish.
https://a360.co/4eL9T1n
Custom shock tolerant mounting to existing UAV frames (Tarot 650).
https://a360.co/4g8TLHZ
Microprocessor & Microcontroller programming using UART/Serial (Navio2 / STM32F1) for flight controller programming and PID Tuning. Used ecalc for verifying TWR/Flight time.

3D-printed RC warships — FFF printing & ingress protection (2017–2019)

A.A.S. graduation project. CAD-to-FFF pipeline for saltwater-resistant scaled warships.
Target: IPX4 (ideally IP67), ~30 min runtime.
Documented FFF layer-adhesion water-ingress limits, post-processing requirements, custom HC-12 RF TX / RX and PWM motor control on ATmega2560.

Fusion 360

MN Aslace (RC-converted, project hull):
https://a360.co/4ctFrYs
VMF Stalingrad (model only, not RC-converted):
https://a360.co/4dYtYkF

MyMiniFactory publications (Alex Kim / @dongk99)

Battlecruiser Stalingrad:
https://www.myminifactory.com/object/3d-print-project-82-battlecruiser-stalinrad-printable-78154
USS Alaska CB-1:
https://www.myminifactory.com/object/3d-print-printable-uss-alaska-cb-1-85035

Education

A.A.S. Electrical Engineering Technology — Hudson County Community College — 2019
B.S. Electrical & Computer Engineering Technology — NJIT — attended through 3rd year, not completed

Technical skills

Thermoelectric integration: module derating, stack assembly, TIM selection, direct-die liquid-metal application, thermal-loop design.
Battery engineering: cell selection, pack topology (series / parallel), BMS programming, thermistor-based thermal management.
CAD & simulation: Fusion 360 (static and transient stress), SolidWorks.
Fabrication: FFF 3D printing, spot-welding CNC (battery assembly), machine-shop practice.
Programming: C++, some VHDL.
Instrumentation: oscilloscope, spectrum analyzer, multimeter, thermistor arrays, HWINFO-based sensor telemetry.

Languages

English (native), Korean (native).

Technical publications

DIY-battery — E-Bike Wiki. Complete PLM (feasibility, component selection, planning, design, assembly) of PEV Li-ion battery. Sole author.
https://ebikewiki.com/index.php/DIY-battery

E-bike battery servicing 101 — Google Forms progressive quiz on Li-ion handling, embedded in the DIY-battery article. Sole author.
https://docs.google.com/forms/d/e/1FAIpQLSd-Jee1HL-Nlk5ZMCbuMsobD1qEIL67Lgq1BAB8IZPk3SZLIQ/viewform